

Fig. 3: Actual size PCB layout of the auto-reversible DC motor control

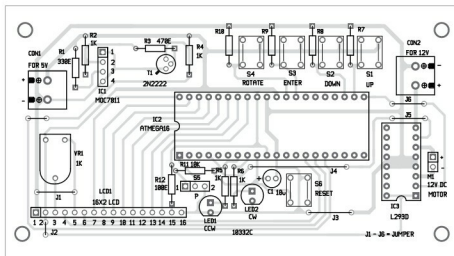


Fig. 4: Components layout of the PCB

SPDT switch S5 is connected to port B pin PB0, so that when it is connected to +5V it gives logic 1 input, and when connected to ground it gives logic 0 input to pin PB0. S6 is used as the reset switch for IC2.

LED1 and LED2 are connected to pins PB6 and PB7 of IC2 for anticlockwise and clockwise indications, respectively.

Circuit operation

The motor is initially in stop position. When power is connected to the circuit, "Set rotations" message is displayed on LCD1. Here, you need to enter the required number of motor rotations by pressing S1 or S2.

These two switches work as up and down arrow keys for the number of rotations and speed, respectively.

S1 is used to increase the number of rotations by five, and S2 is used to decrease the number of rotations by five. The minimum number of rotations is ten.

Once the desired number of rotations is selected, press Enter (S3). The LCD display will show "Rotations set

to xx" message. After two seconds, you will be prompted to enter the required speed, when the LCD displays "Set motor speed" message.

You can set the speed from 10 to 100 per cent in steps of 10 by pressing up or down arrow keys. Once the desired speed is set, press Enter. The display now shows "Speed is set to xx%" message.

Now, you will be prompted to start motor M1 through "Press rotate" message on LCD1. When you press the rotate button (S4), microcontroller IC2 starts applying PWM on PD7, as per the selected speed, and the motor starts rotating anticlockwise.

LED1 turns on to indicate that the motor is rotating in anticlockwise direction. PWM width is increased to the desired speed in two or three steps, so that the motor speed increases step by step.

As the motor rotates one revolution, a light thin metal strip attached to the shaft of the motor passes through the gap of the opto-interrupter sensor module. This interrupts IR light falling on the phototransistor for a fraction of a second. The phototransistor gives a very short-duration positive pulse, which is inverted by transistor T1, and it gives a short-duration negative pulse to pin PD2 of IC2.

This negative pulse is registered as one count by IC2. In this way, IC2 counts the number of interrupts as the number of rotations of the motor and displays it on LCD1.

As the motor reaches the last six to ten rotations, PWM width is reduced to decrease the speed. For the last two to four rotations of the motor, its speed is reduced to 10 per cent to further slow down the motor.

As the motor completes the rotations in anticlockwise direction, PWM is generated from PB3 pin. The motor starts rotating in clockwise direction. LED2 turns on. Again, motor speed is increased in steps to full speed and then decreased in steps to the mini-

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com